



# ENCLOSURE

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## **Sprig Enclosure Guide:**

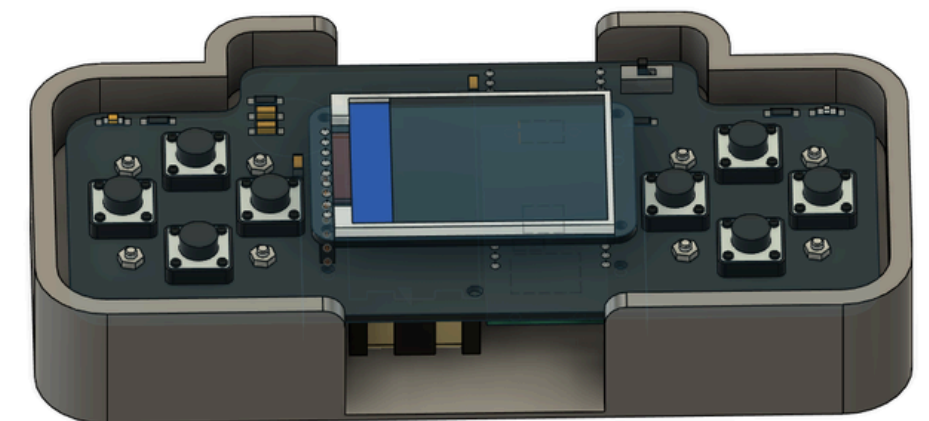
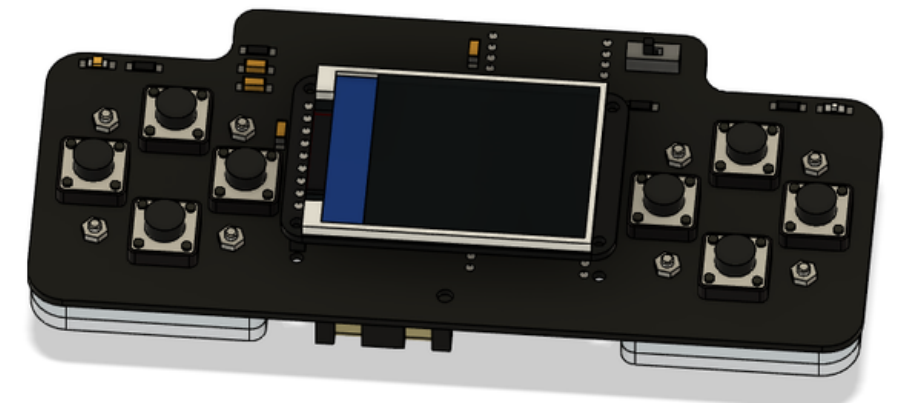
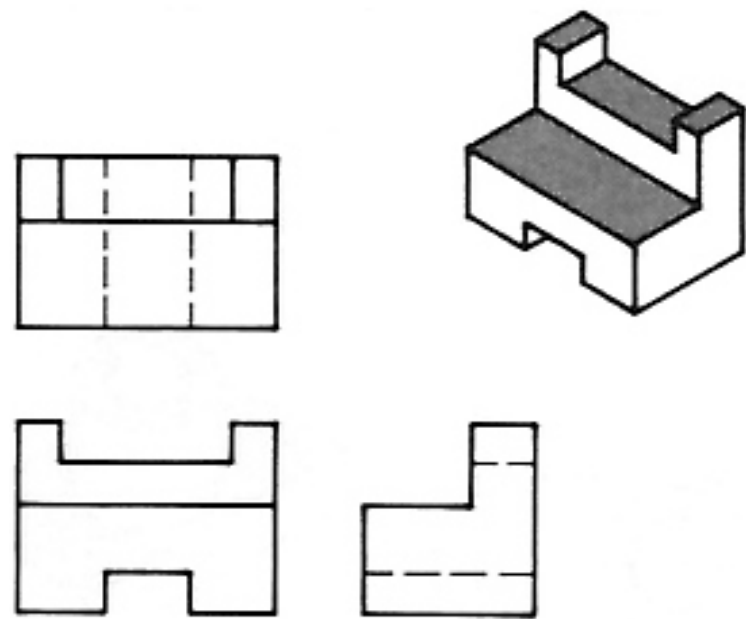
Design a custom enclosure for your project, get a 3D print of it shipped to you, and win other prizes!





# Objectives

1. **Analyzing Drawings:** Learn how to interpret and analyze simple orthographic drawings
2. **Downloading Models:** Download the fusion file for the sprig
3. **Understand Tolerance:** Review the importance of tolerance and how/when to incorporate it into our designs
4. **Hands-on Practice:** Follow steps to design a basic enclosure for the Sprig
5. **Custom Design:** Apply skills to design a unique CAD enclosure from scratch





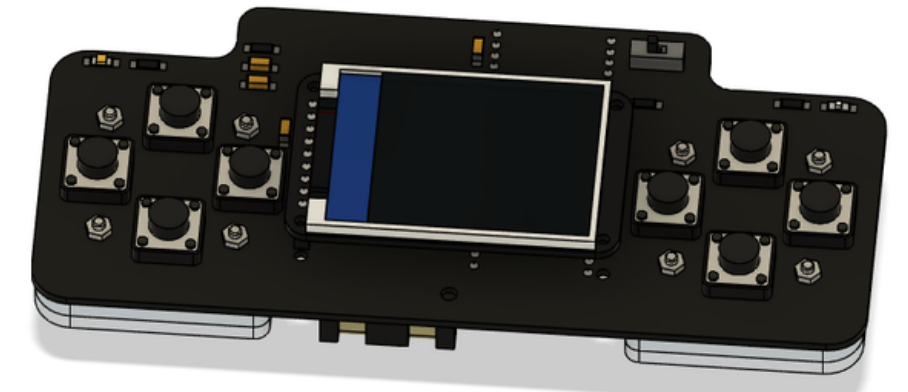
# Sprig 3MF

We're glad to say that we developed a relatively accurate 1-1 model of the sprig board for you to import into your model and test to make sure your finished enclosure is practical!

Make sure to download the FULL 3mf file.

Download it at this link:

[https://drive.google.com/file/d/1sl9yy5y4U7\\_sG343yVt3JUDSh5CydAHA/view?usp=sharing](https://drive.google.com/file/d/1sl9yy5y4U7_sG343yVt3JUDSh5CydAHA/view?usp=sharing)



# Orthographic Drawings

## **This Project Teaches:**

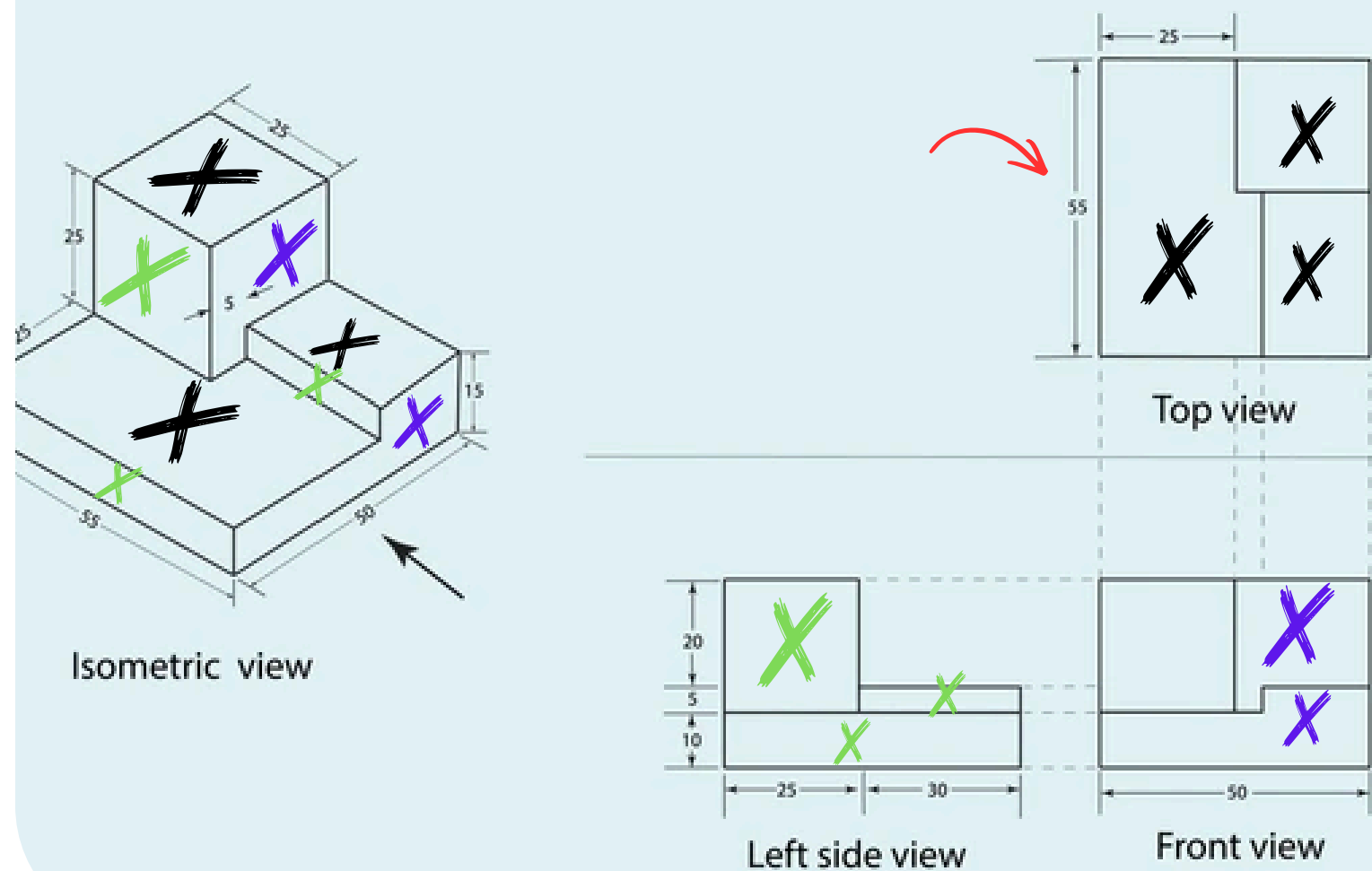
1. Basic understanding of orthographic drawings
2. Introduction and display of the sprig





# Orthographic Drawings

## ORTHOGRAPHIC PROJECTIONS

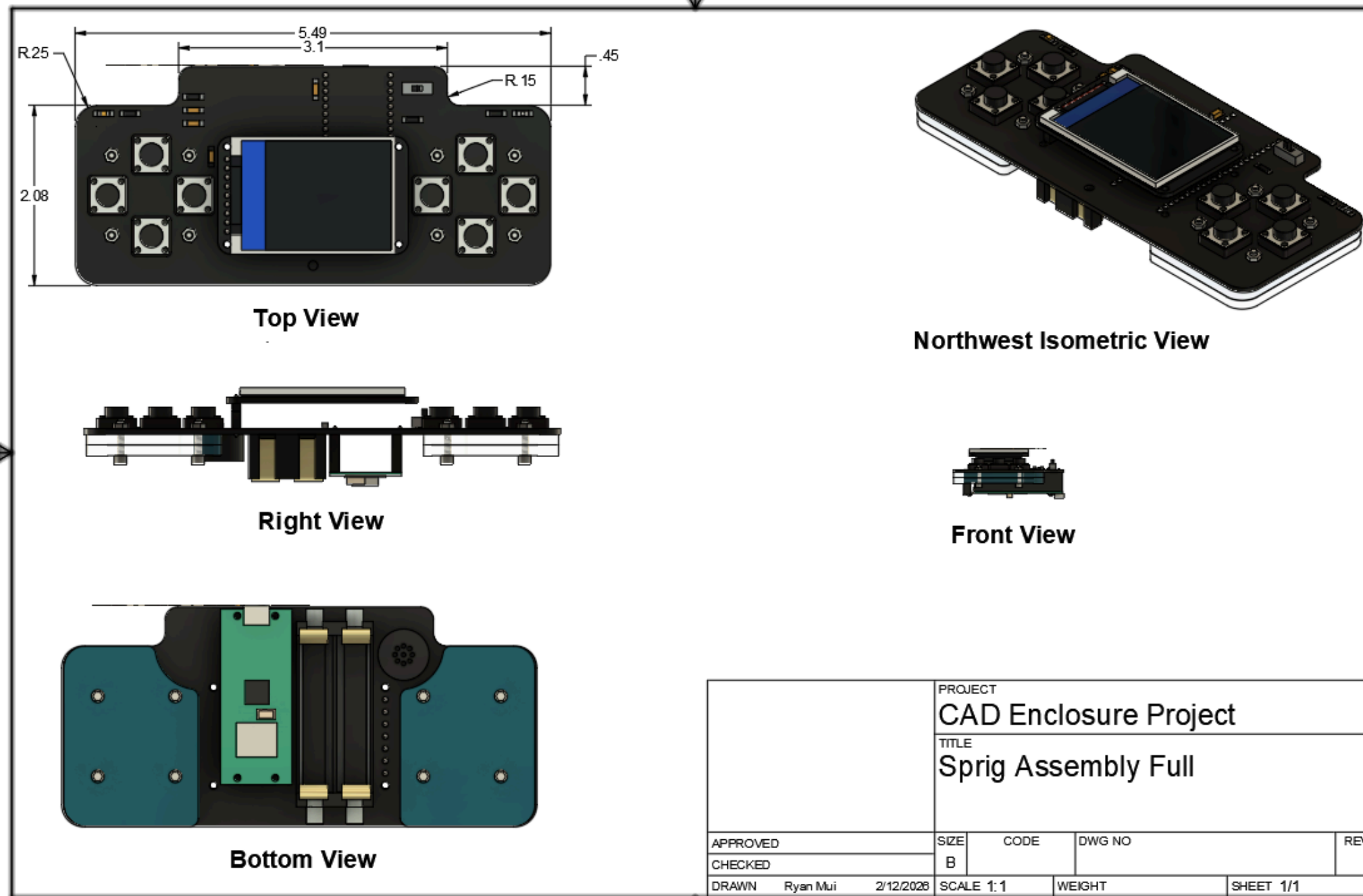


Orthographic drawings are technical drawings that represent a three-dimensional object using multiple two-dimensional views, each drawn from a different angle. The three-dimensional isometric view is split into 3 two-dimensional views: Top, Side, and Front views. The colored x's in the model will help you correspond which faces are in which views. As you can tell, orthographic drawings also allow us to see the dimensions of the figure. The dashed lines also help us see the different views in reference to each other.





# Sprig Showcase



Looking at the drawing, we can see that the sprig has many intricate components, all of which have dimensions that can be accounted for. If we were to use an orthographic drawing, it would get cluttered pretty quickly. Thankfully, fusion has a dimension tool, which we will be using later to measure the smaller details. We will be using the drawing to make the base plate, though.

# Assessing Designs

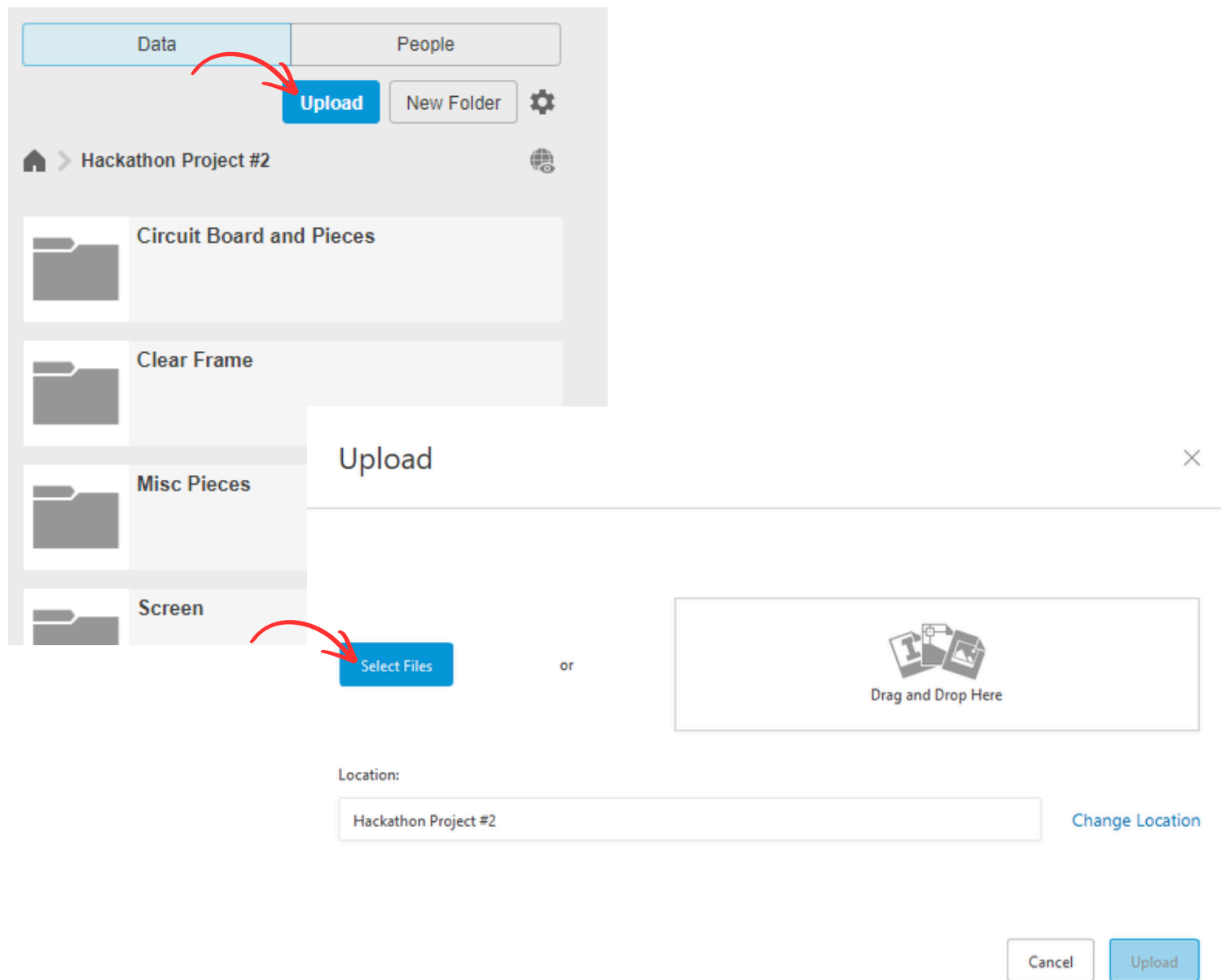
## **This Project Teaches:**

1. How to download and upload  
Fusion designs
2. Adjusting and optimizing  
visuals
3. How to measure and analyze  
the dimensions of designs





# Downloading Models

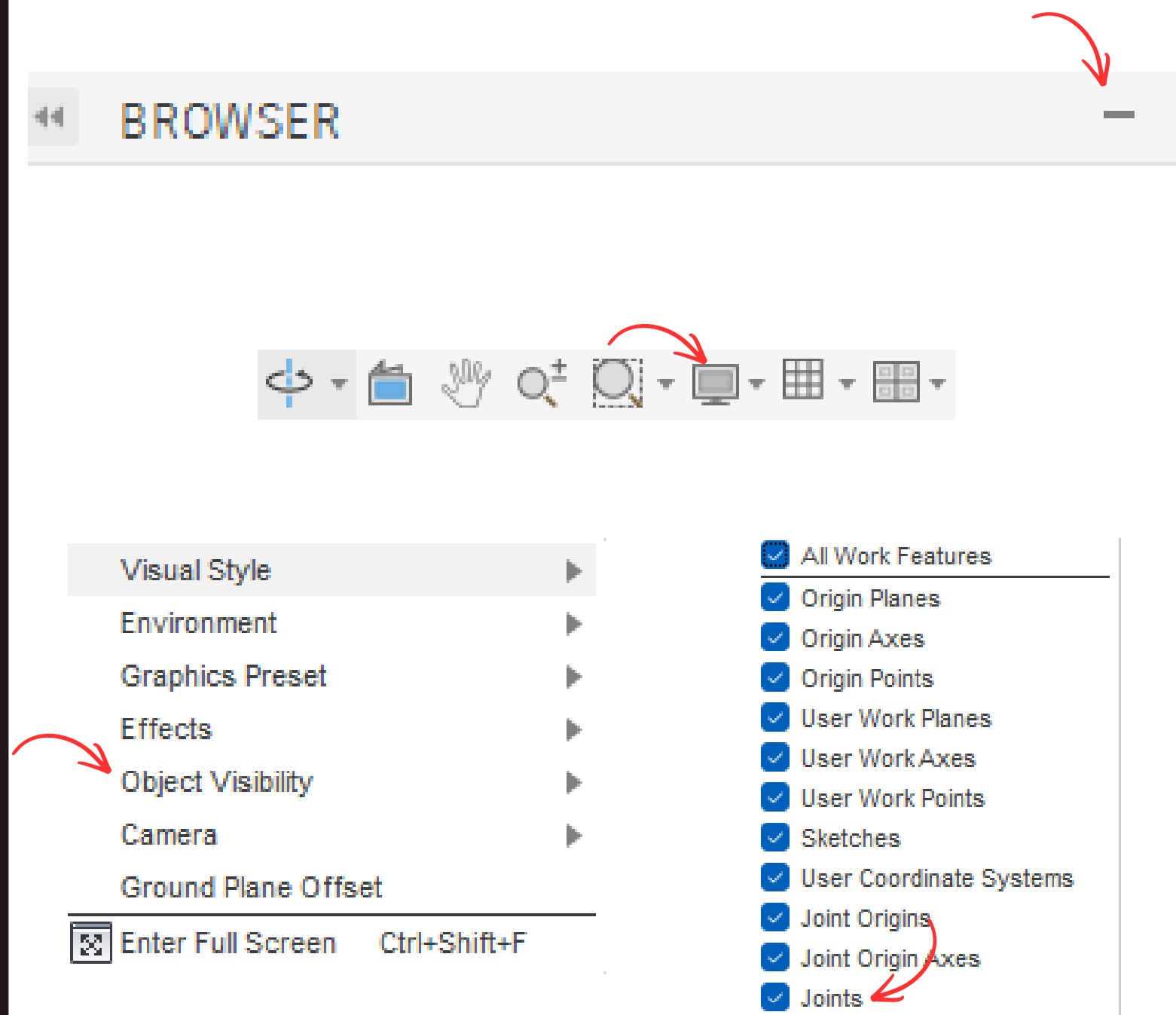


Let's get started by first downloading a model of the sprig, which we can use for a reference. Make sure you downloaded the provided 3MF file for the sprig first. Make a new project, and at the top left corner of your screen, click the blue upload button. A menu will appear, and hit "select files". We can add the sprig file and upload it to the project. The sprig assembly should now appear on the left-side menu. Double click it and it will open on your main screen.





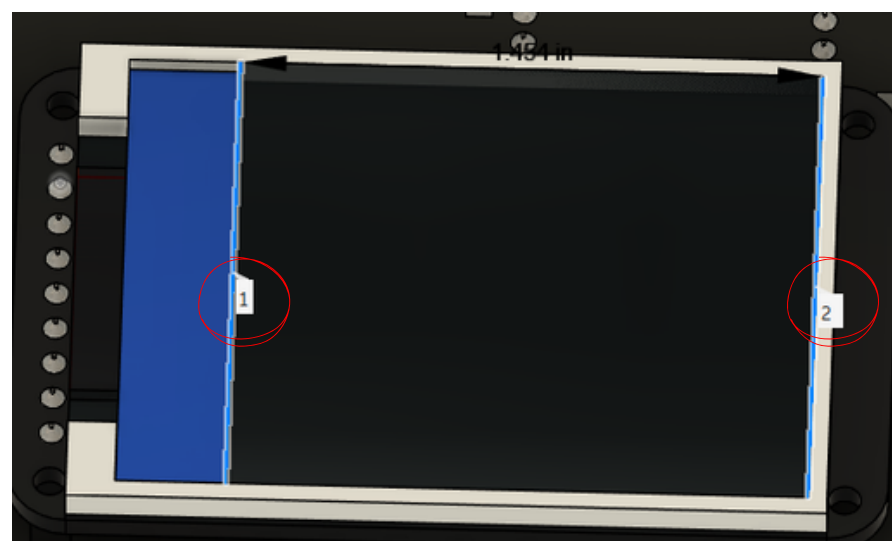
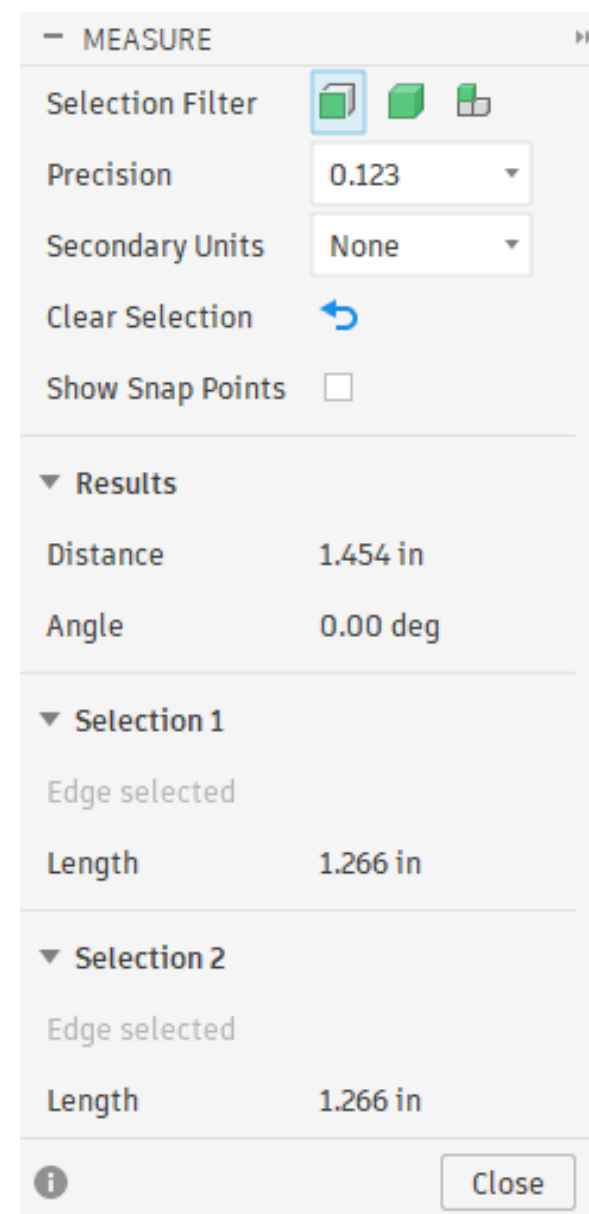
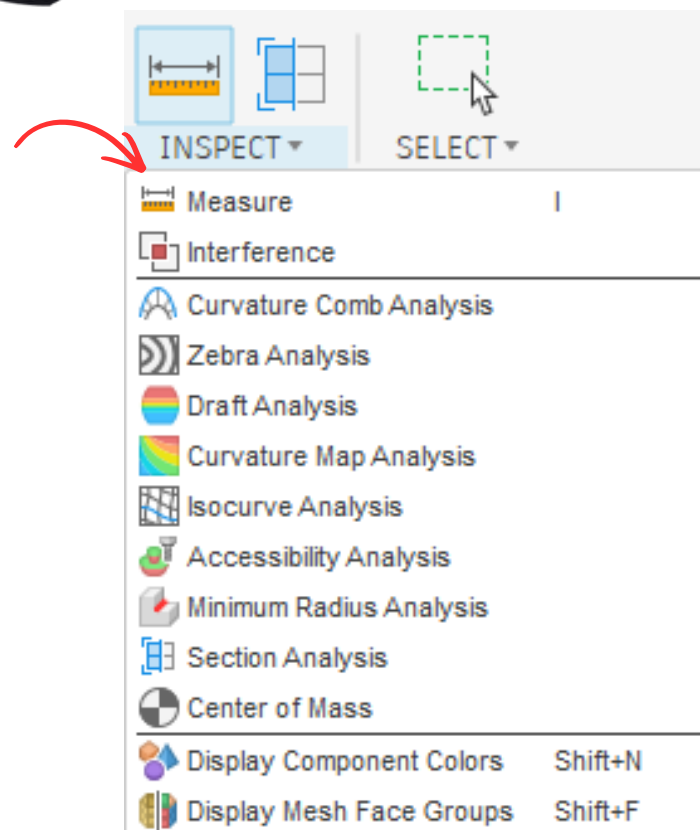
# Adjusting Visuals



As you can tell, the design is currently very cluttered due to all the joints used to put all the components together. Obviously, this would be a hassle to work with so let's change the visuals on our model. At the bottom of the screen, select the display settings icon as shown on the left. In the pop-up, open the object visibility menu and deselect joints. For clearer visuals, you can also turn off sketches; however, you will need to turn them back on again when we start designing the enclosure. In the top left corner, also minimize the browser to remove more of the clutter off your screen.



# Measuring Designs



Before we make our enclosure, let's take a look at how to measure our designs. In the top right corner, under the Inspect menu, select the measure tool, or shortcut I. The dimension tool allows us to measure individual sides and the distance between different pieces. The "1" and "2" correspond to the dimensions of the selected sides as shown on the right side menu. This tool can be used not just for linear sides, but for all shapes.

# Review of Tolerance

## **This Project Teaches:**

1. What tolerance is and why it matters
2. Types of tolerances
3. How tolerance affects assembly and usability



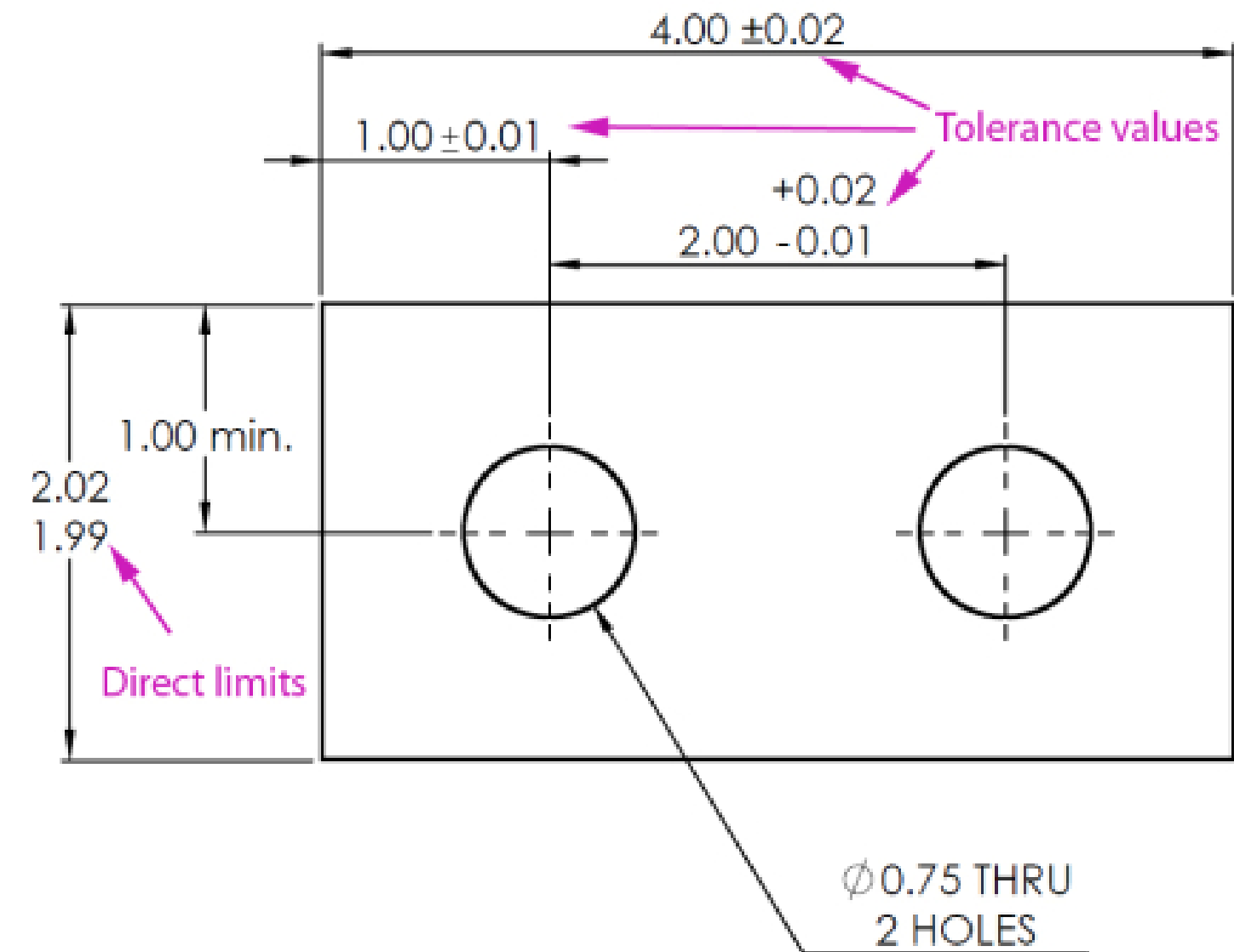


# Definition of Tolerance

Now before we move on, let's quickly recap the importance of tolerance.

**Tolerance** is the allowable limit of variation in a physical measurement or characteristic of a part.

Because no manufactured part can be made perfectly, tolerance tells builders how much deviation is acceptable while still allowing the part to function as intended.





# Different Types of Tolerance Fits

1

A **clearance fit** always leaves space between the shaft and the hole, even at the tightest tolerance limits. They're usually used when low friction is needed, or if easy assembly/disassembly is required.

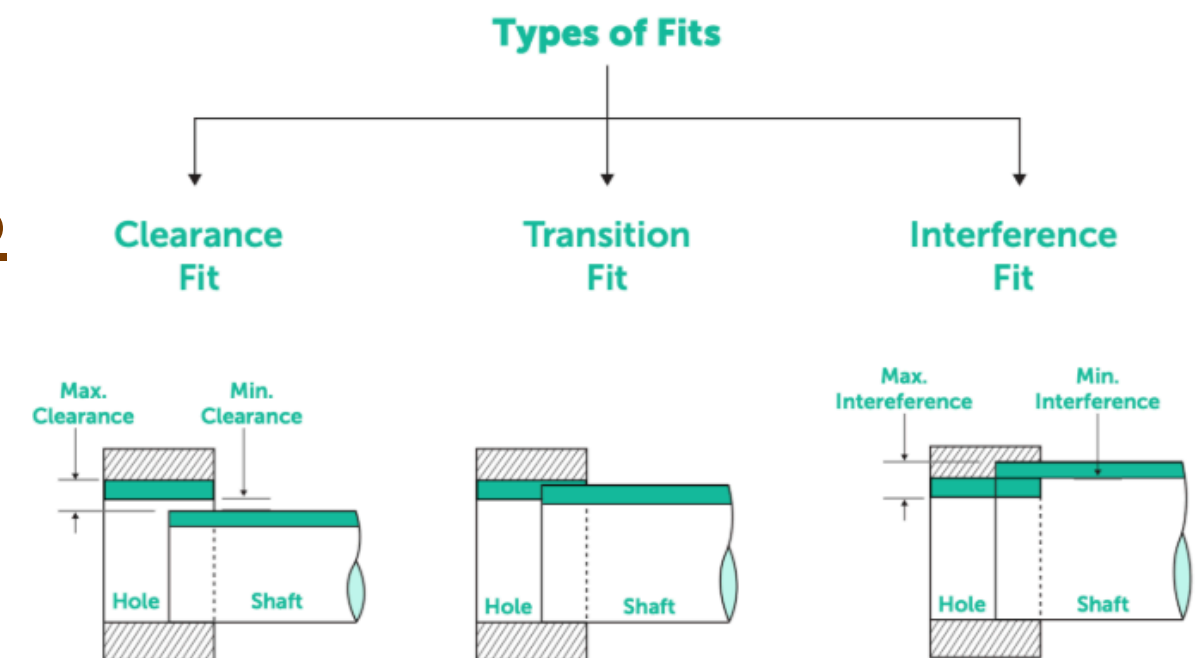
2

An **interference fit** means the shaft is larger than the hole at all tolerance limits. They are usually used for creating tight or permanent joints.

3

A **transition fit** can result in either a small clearance or small interference, depending on the tolerance limit. They are often used for creating accurate fits when components need to align precisely.

Our enclosure will predominantly be using clearance fits so we can easily remove the sprig.





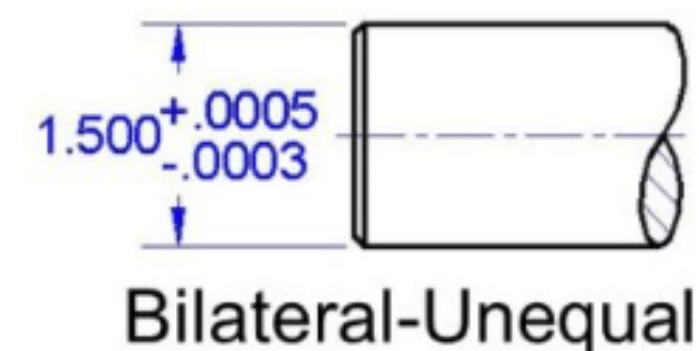
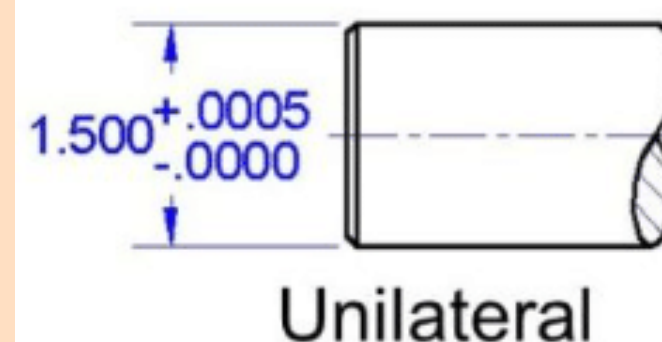
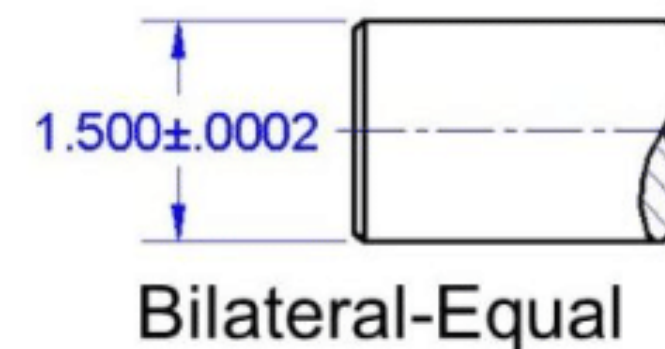
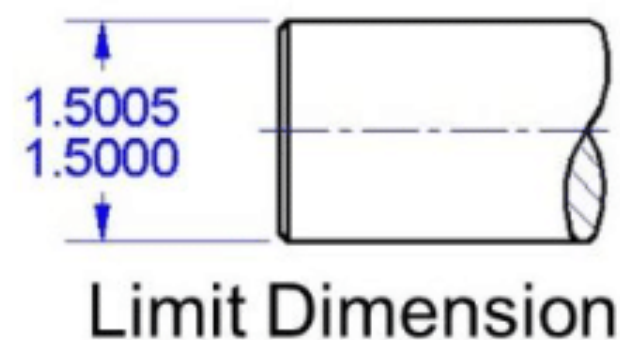


# Unilateral vs Bilateral Tolerance

**Unilateral tolerance** varies in only one direction. They can vary in either the positive or negative direction, but not both.

**Bilateral tolerance** varies in both directions. Bilateral tolerance is usually indicated by a  $\pm$  sign; however, unequal bilateral tolerances can exist, as shown on the right. Limit dimensions specify the maximum and minimum allowable sizes directly (e.g., 1.98–2.02 in), defining the full acceptable range.

Since we are designing a custom enclosure, there are no provided tolerance values. We will be using a bilateral tolerance of 0.08" for all values





# Designing the Enclosure

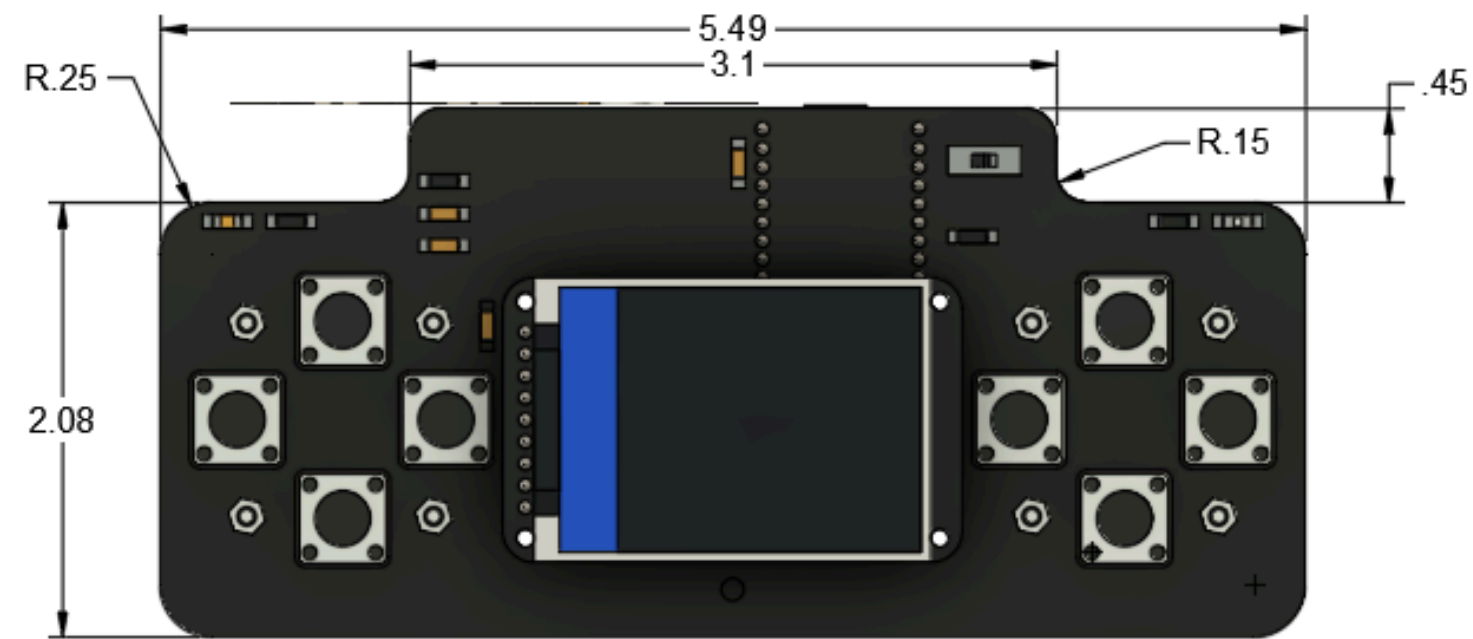
## **This Project Gives Practice In:**

- Analyzing and following orthographic drawings
- Applying our understanding of tolerance to real-world designs
- Using our acquired knowledge and skills of Fusion 360
- Preparation for our own custom models

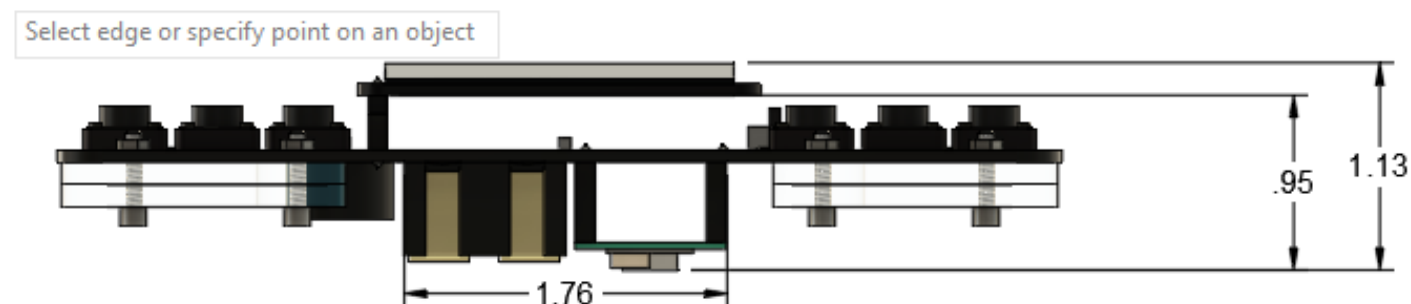




# Referencing Drawings



Top View

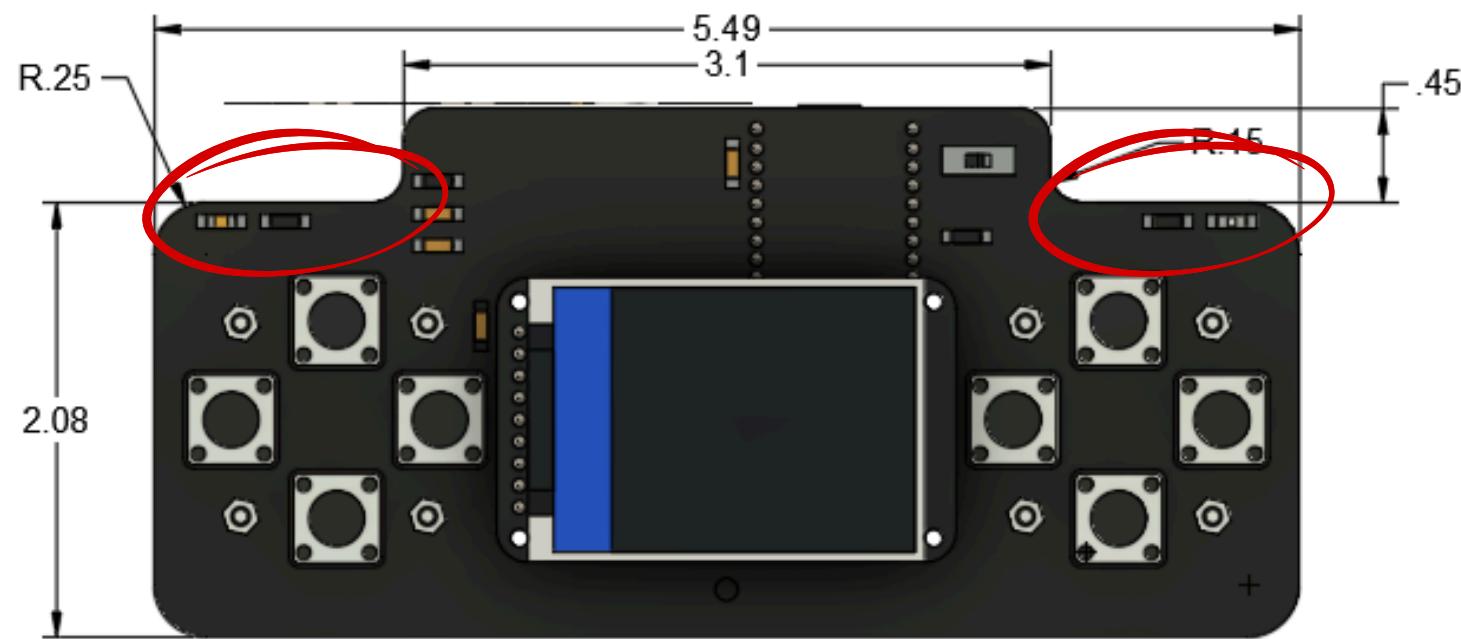


Right View

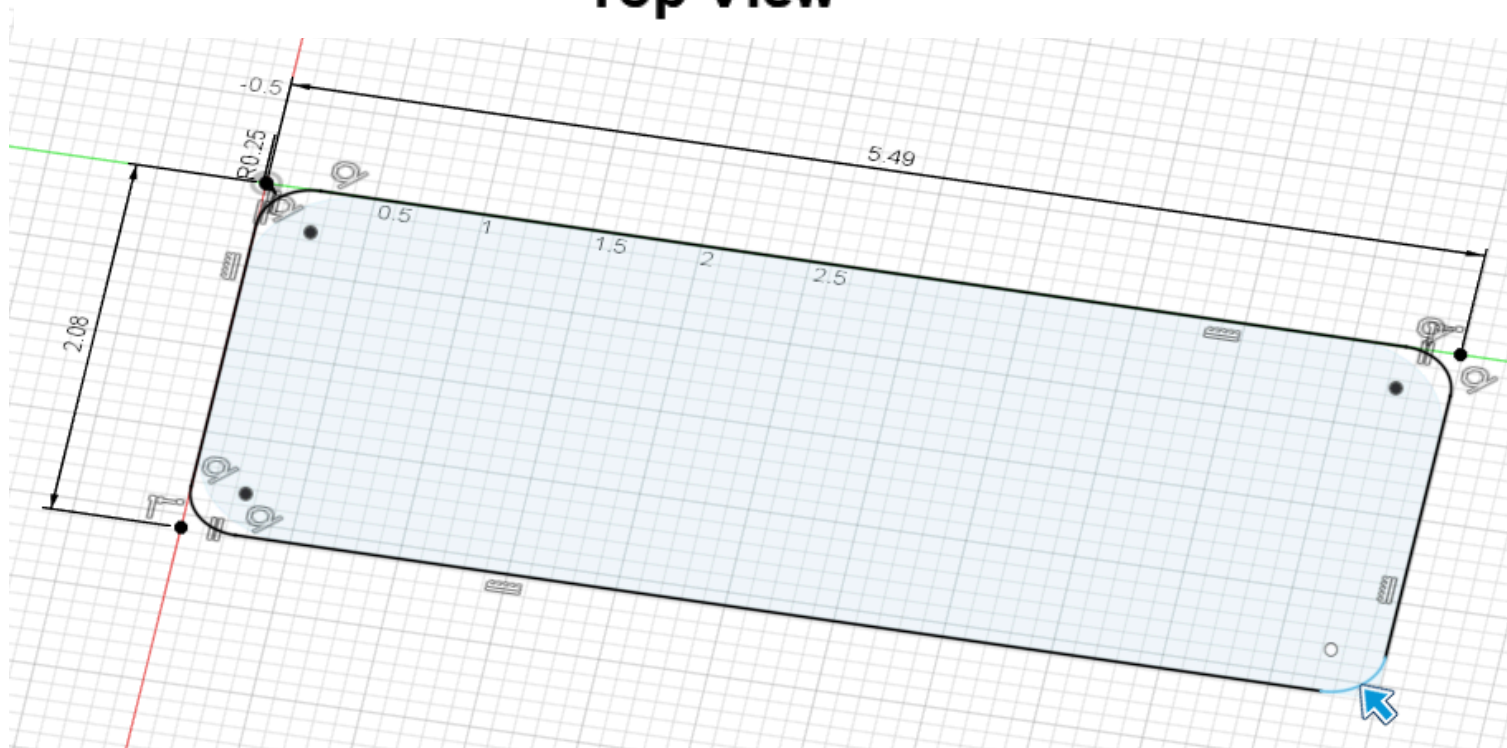
Although we *can* use the dimension tool instead, analyzing orthographic drawings is generally a helpful skill to have. Since the provided dimensions are larger, we can use them to sketch the main base for our enclosure. To start, open up a new design and make a new sketch. Make sure to come back to this slide as we will be using the dimensions for the upcoming set of directions.



# Floor Layout



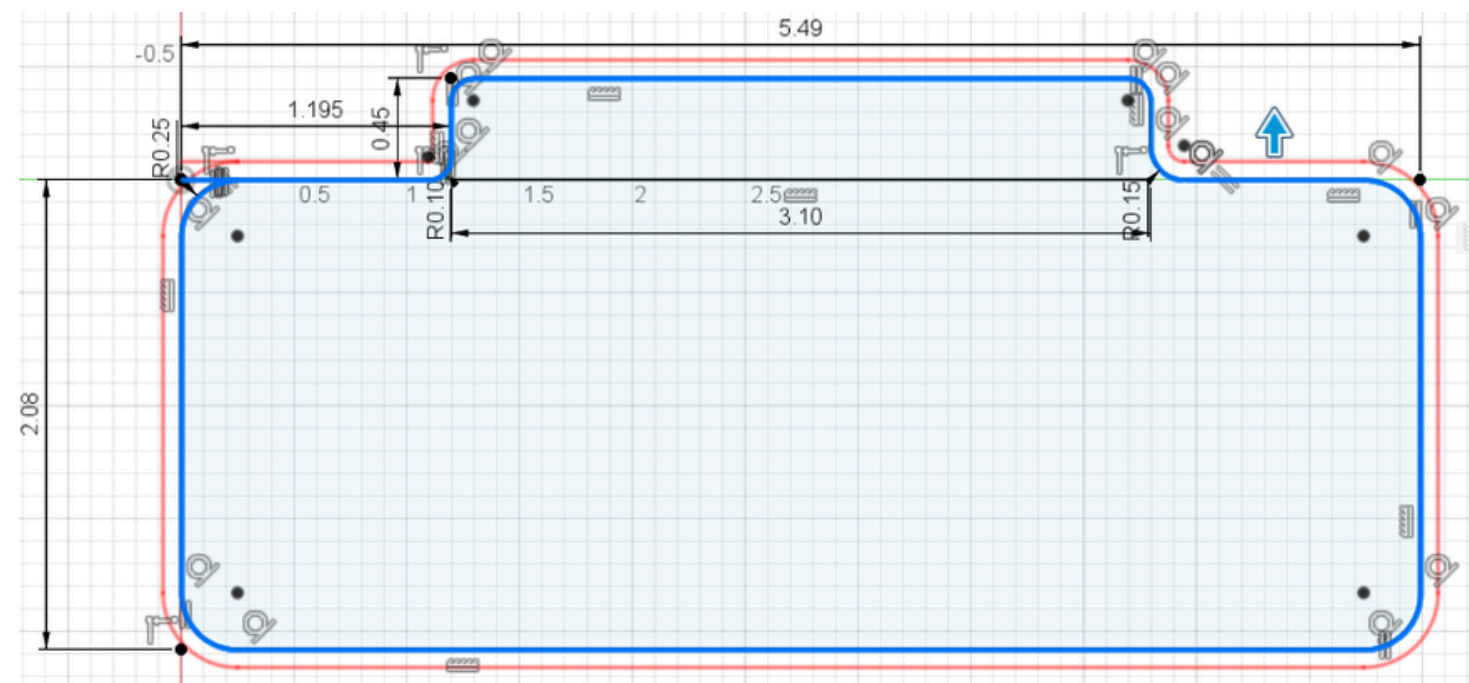
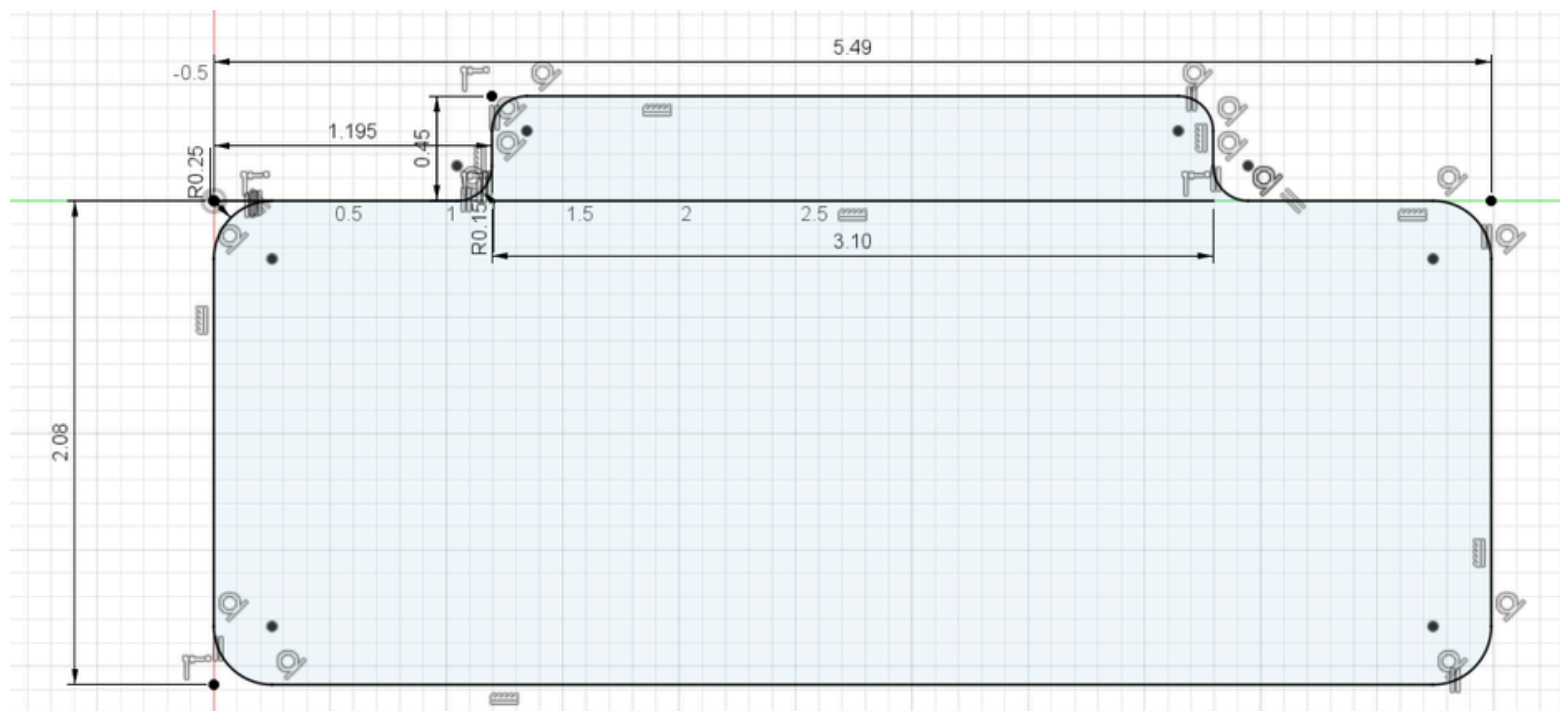
Top View



Looking at the model, we can see that the base of the sprig is made of two rectangles with curved corners. Let's select the XZ plane and sketch out the bottom, 2.08" x 5.49" rectangle. Let's also fillet each corner by .25 in, as shown in our drawing. Since we have the length of our top rectangle, we can calculate the distance between the sides of the top and bottom rectangles (as shown by the circles).  $0.5(5.49-3.1)$  will get us the value between the two rectangles. We multiply by one-half to account for the value being applied on both sides



# Floor Layout Pt.2

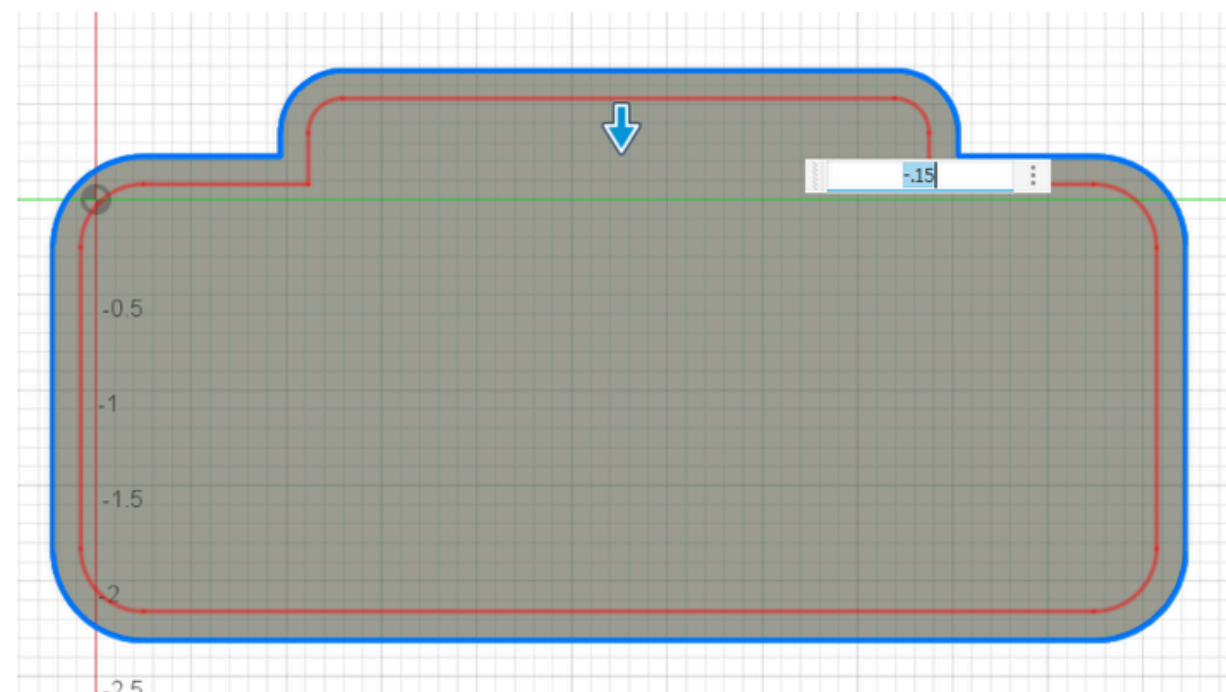
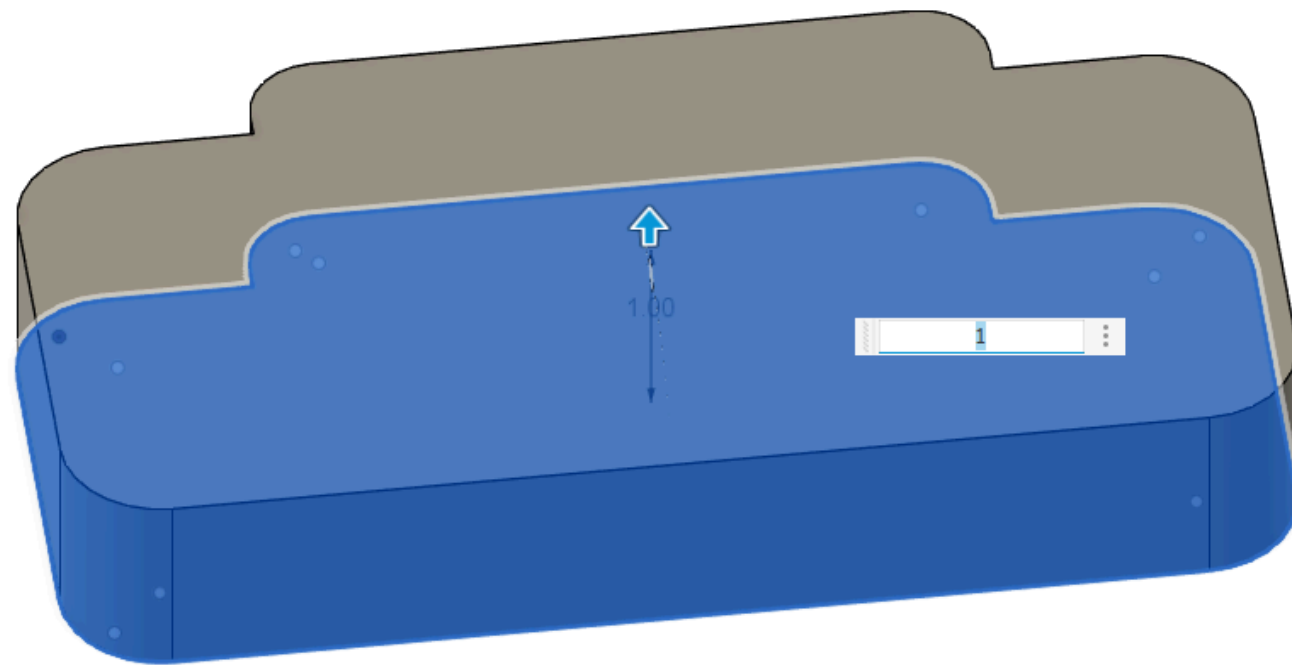


Using our calculated value (1.195"), we can create a placeholder line either from the left or right corner of the bottom rectangle. From the endpoint, let's create the top, 3.1" x 0.45" rectangle and fillet the corners by 0.15". We can now trim the placeholder line. Since we are making a component to fit the sprig inside, the floor will have to be larger than the sprig. Using the offset tool, shortcut "O", apply a 0.23" offset on the current model and use the trim tool, shortcut "t" to remove the excess inner lines afterward. We can now finish our sketch.





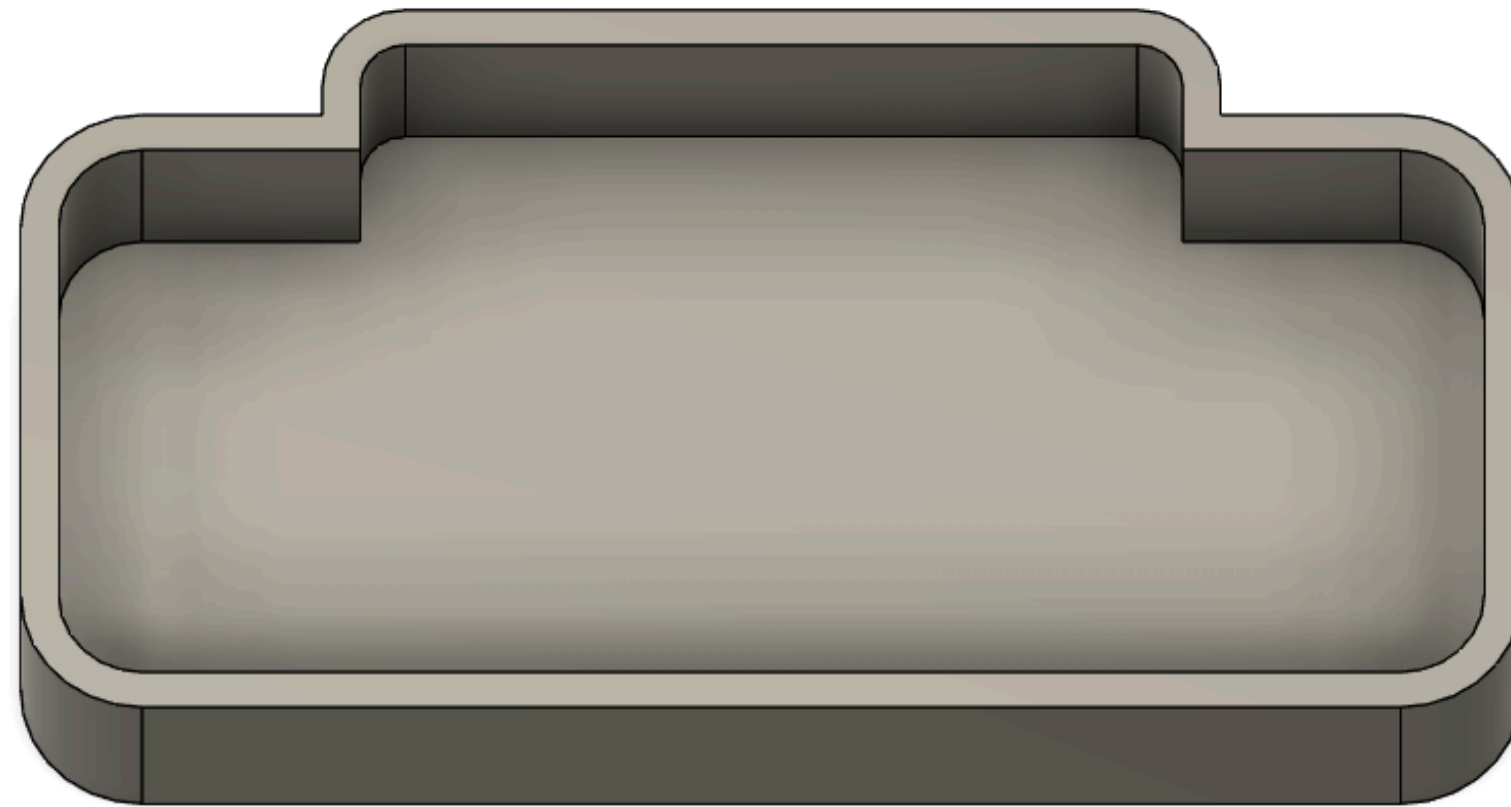
# Enclosure Base



Now we have the basic floor model for our enclosure. Let's extrude the base by 1". We need to make a cutout for the sprig, so select the top face and make an inward offset of 0.15". Notice how we are doing an inward offset of 0.15" since we are adding an extra 0.8" inside to account for the design's tolerance. Now we can finish the sketch and extrude the inner face downward by 0.95".



# Progress Check #1

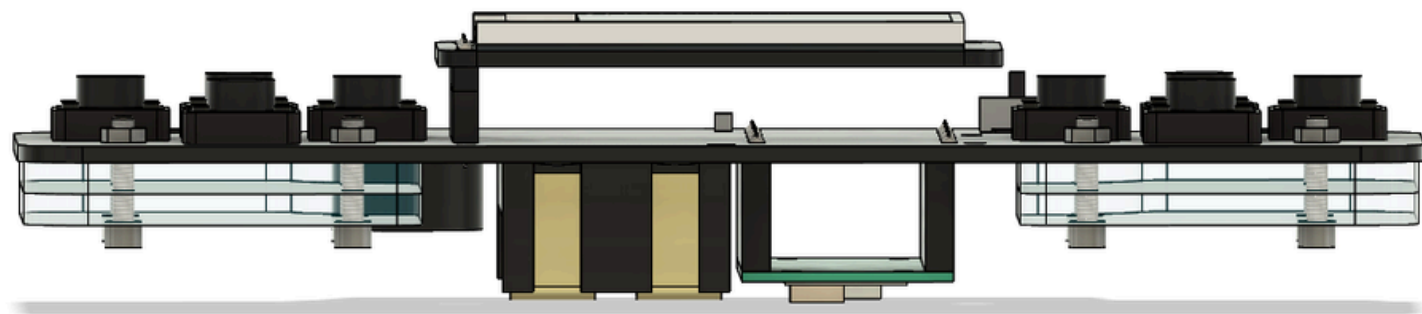
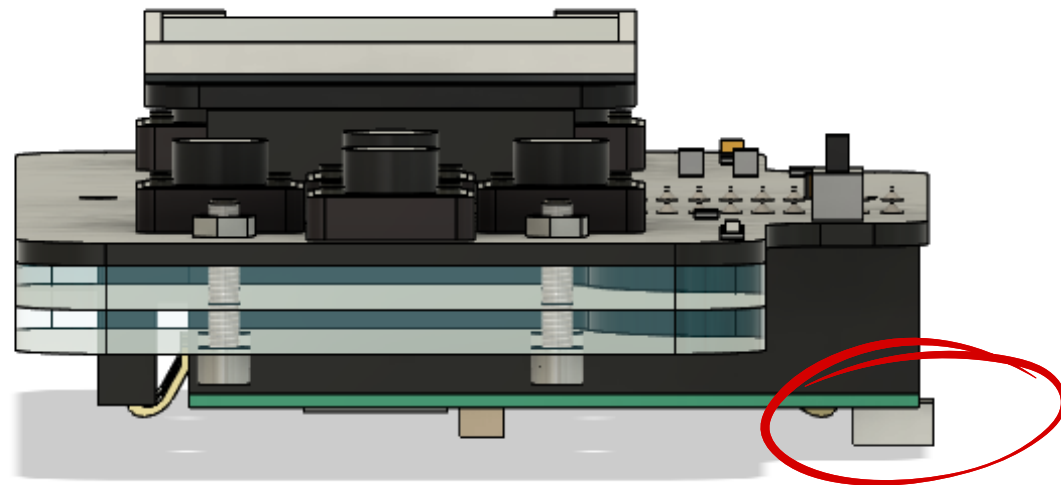


Your model should look like so. If not, look through the previous slides.





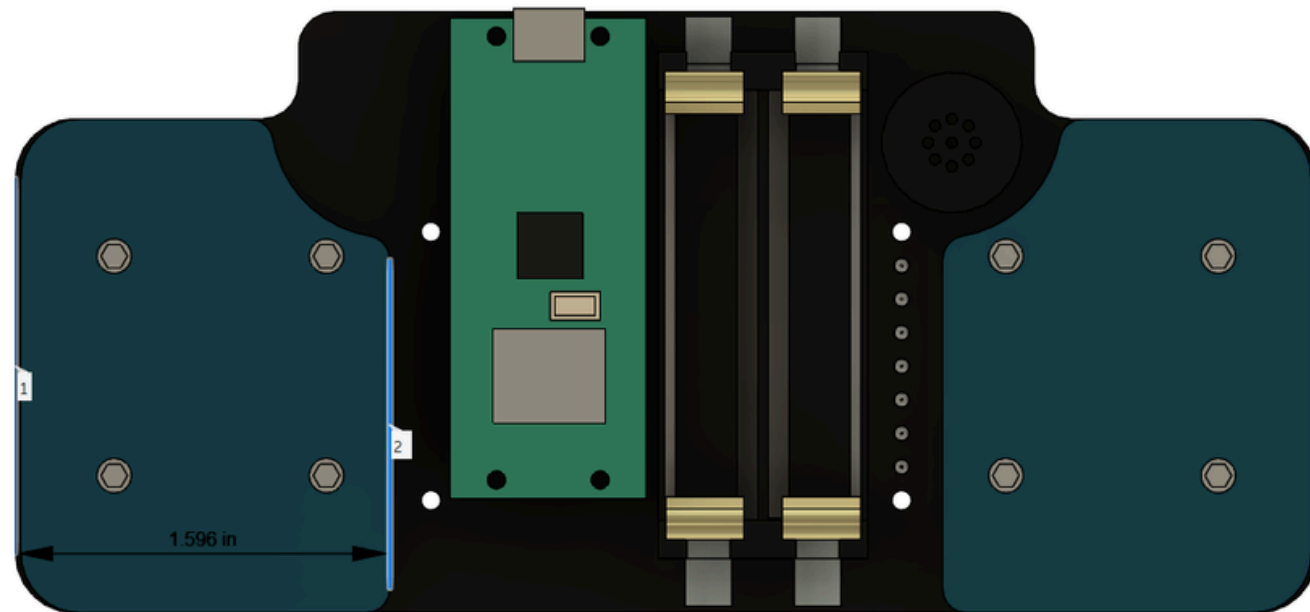
# Assessing Details



Let's take a look at some of the small components we will need to account for. For example, look at the circled port on the left that's attached to the circuit board. Notice how it extrudes over the edge of the sprig's base? We will need to make a cutout so that the sprig can fit inside our enclosure. Additionally, the underside of the sprig has many components that would make placing it inside very uneven. We will also have to level the inside of our enclosure to stabilize the sprig.

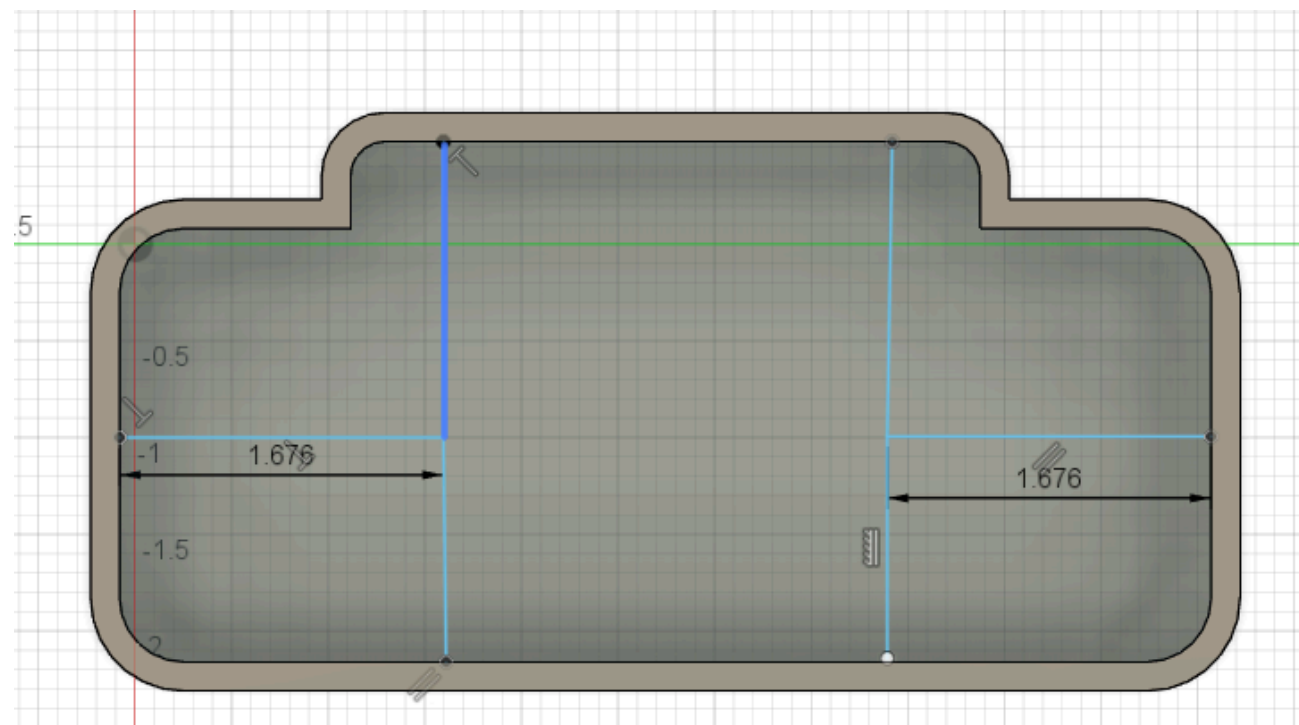


# Stabilizing the Sprig Pt.1



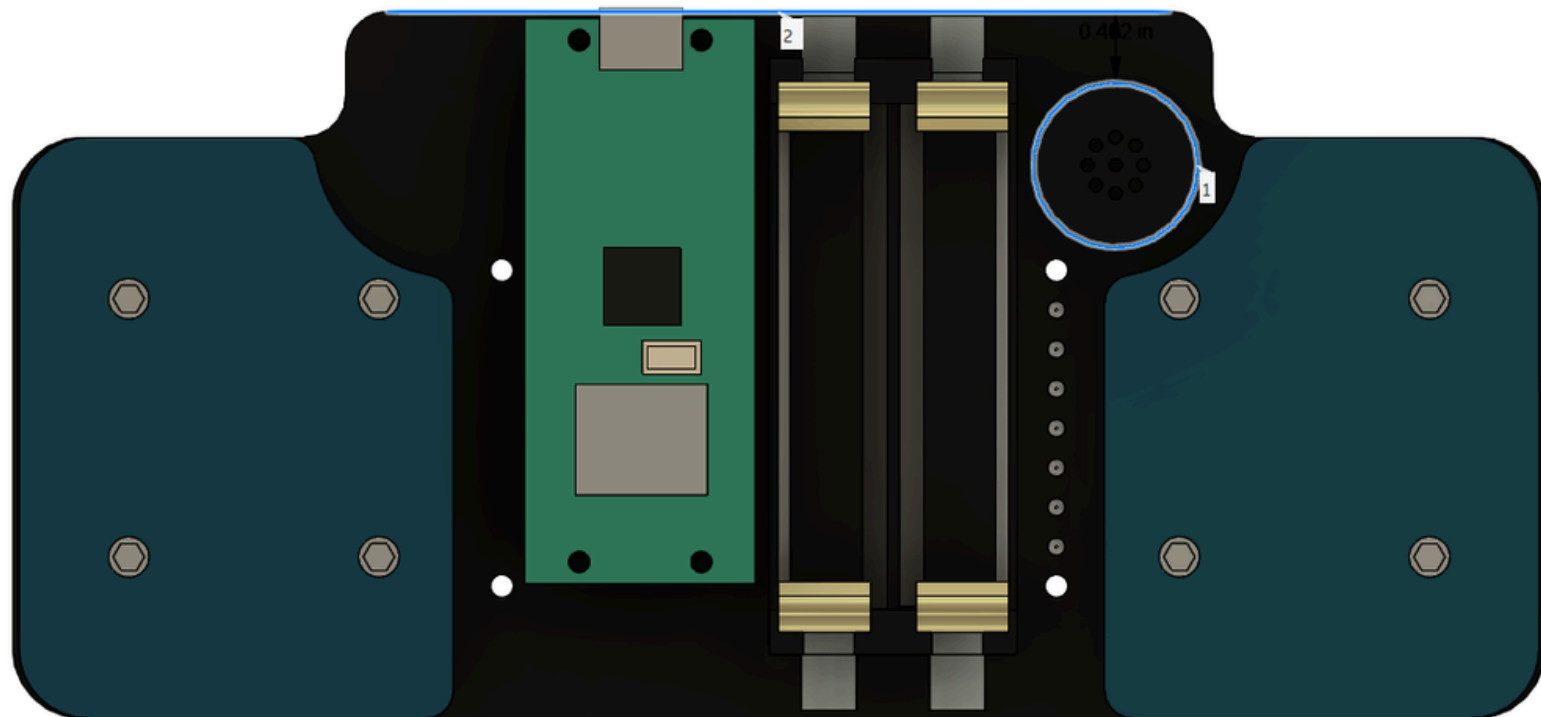
We'll start by making the adjustments to the base to stabilize the sprig. As shown on the left, let's use our measure tool to find the length of the glass panels. Now with our measurement, Let's take our measurement of 1.596" and add another 0.8" to account for the extra room we have made to account for tolerance. Following the bottom picture on the left, section out the center of our enclosure using the measurements we obtained.

You can now trim the placeholder horizontal lines. ***Make sure you can highlight/select each section.***





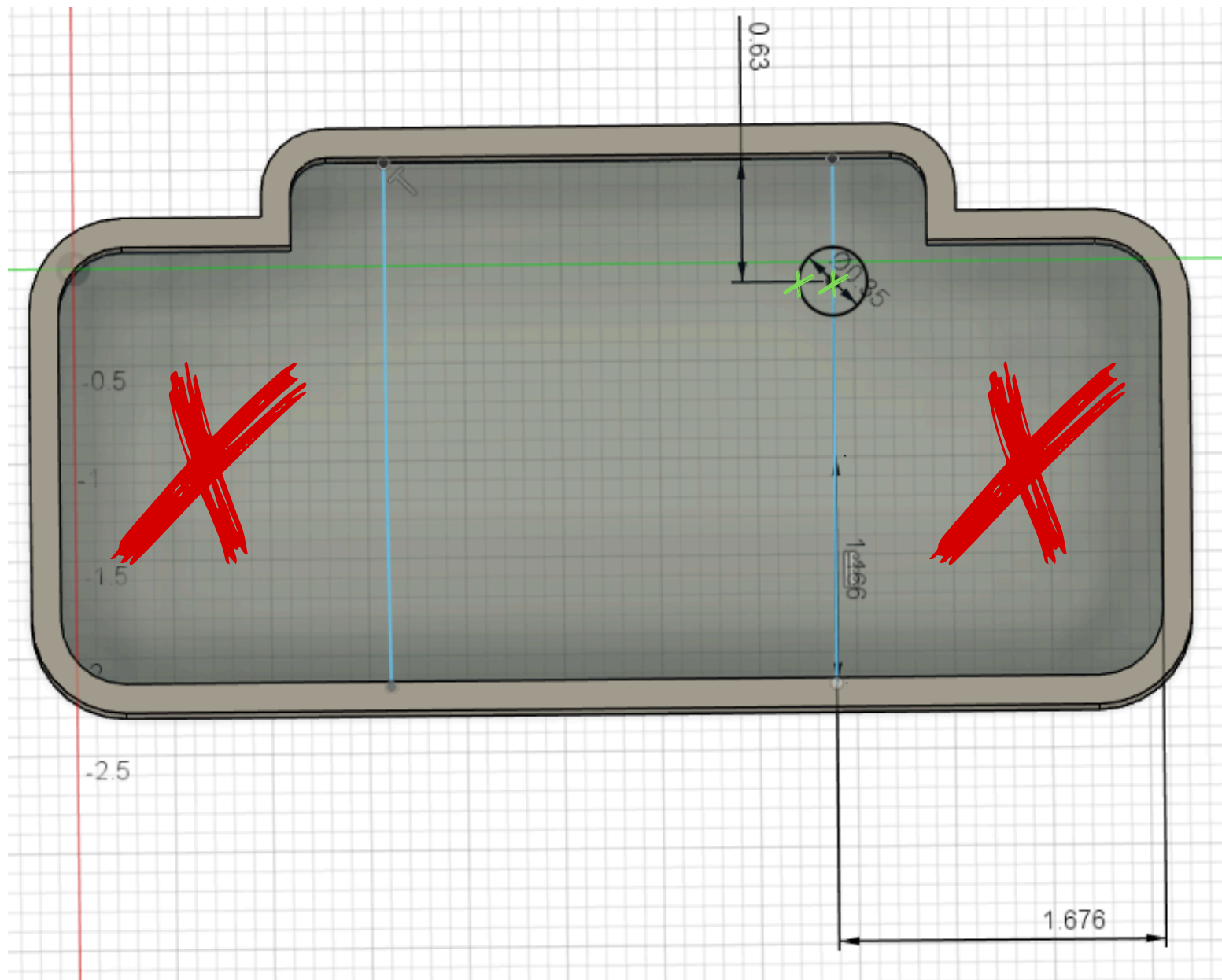
# Stabilizing the Sprig Pt.2



Unfortunately, we also need to account for the circular component on the sprig. Our dimension tool can give us the radius of the circle, and we can find its position on the sprig. The distance of our width from the center will actually be the same as the measurement for our panel earlier (1.676"). Remember that we are measuring from the edge of the circle, so to find our center point's location on the sprig, we need to add the radius of the circle as well for the distance from the length. We also have to add another 0.08" just like before.



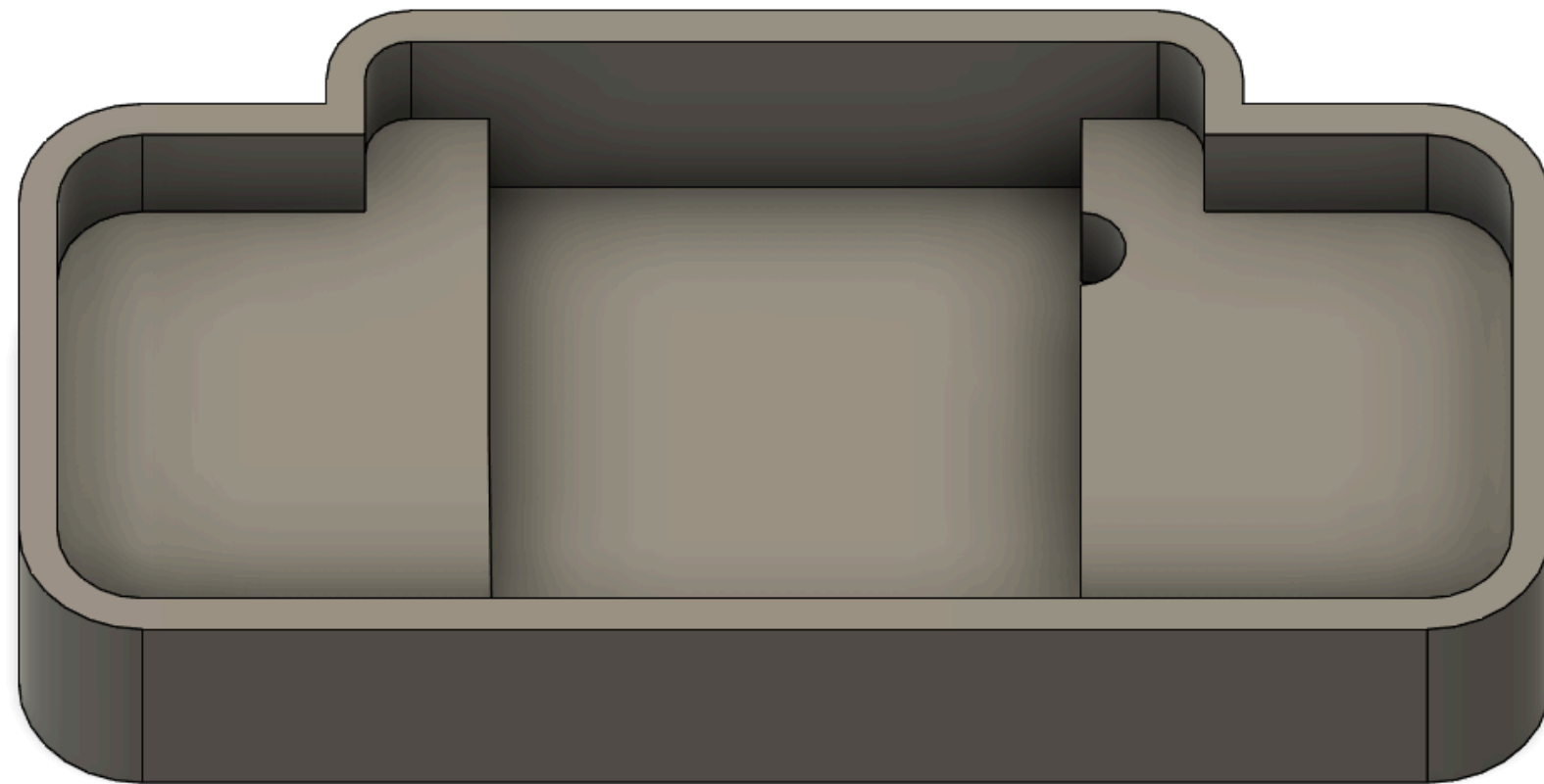
# Stabilizing the Sprig Pt.3



To save some tedious work, on the left is the position of the circle. We enlarged the radius of our circle to account for tolerance once again. The green x's mark the lines that we can then trim. Now, the faces marked by the red x's, let's extrude those by 0.45".



# Progress Check #2

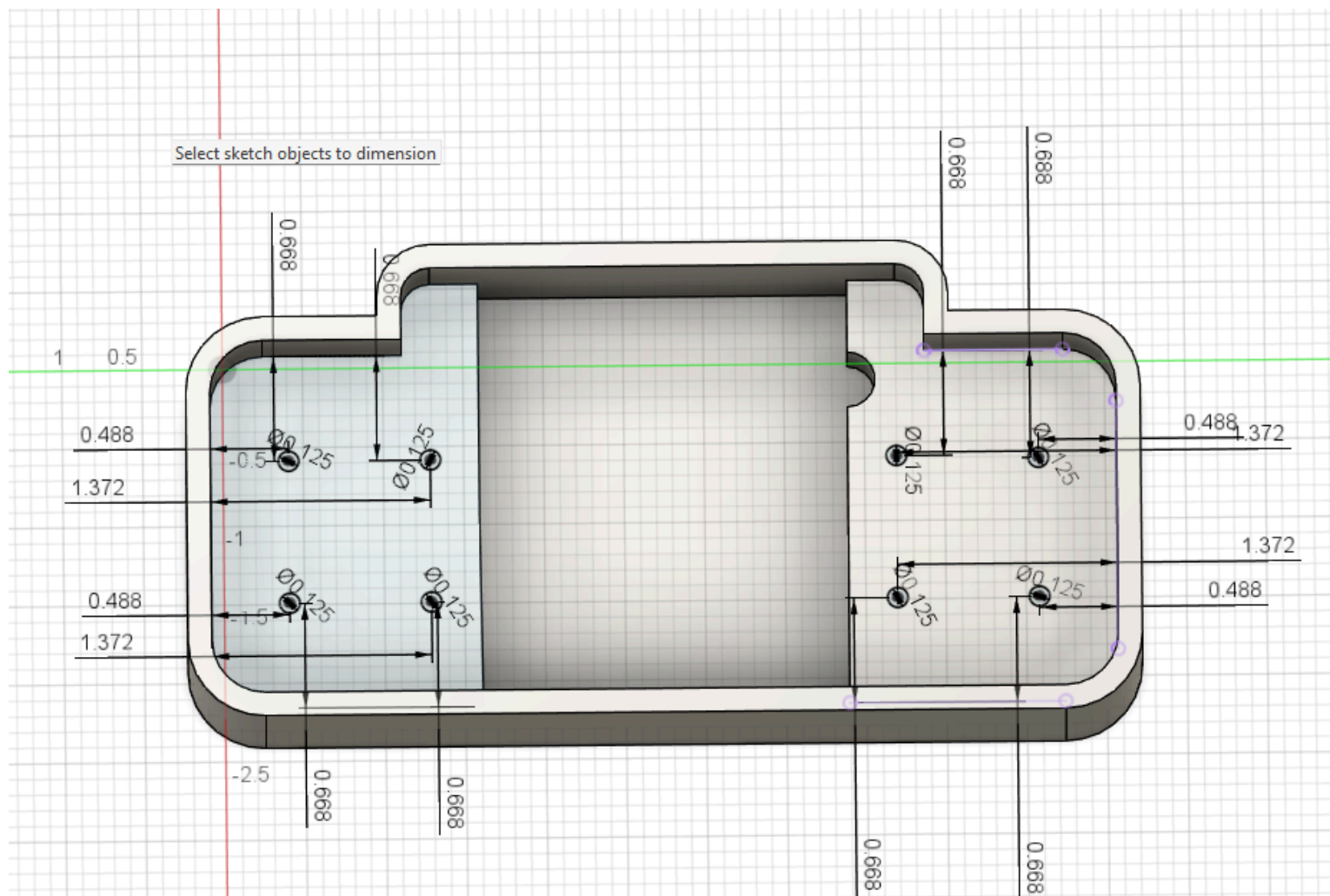


If you're with us so far, nice job. With a design as complicated as the sprig, there are a lot of meticulous details we need to account for.





# Stabilizing the Sprig Pt.4

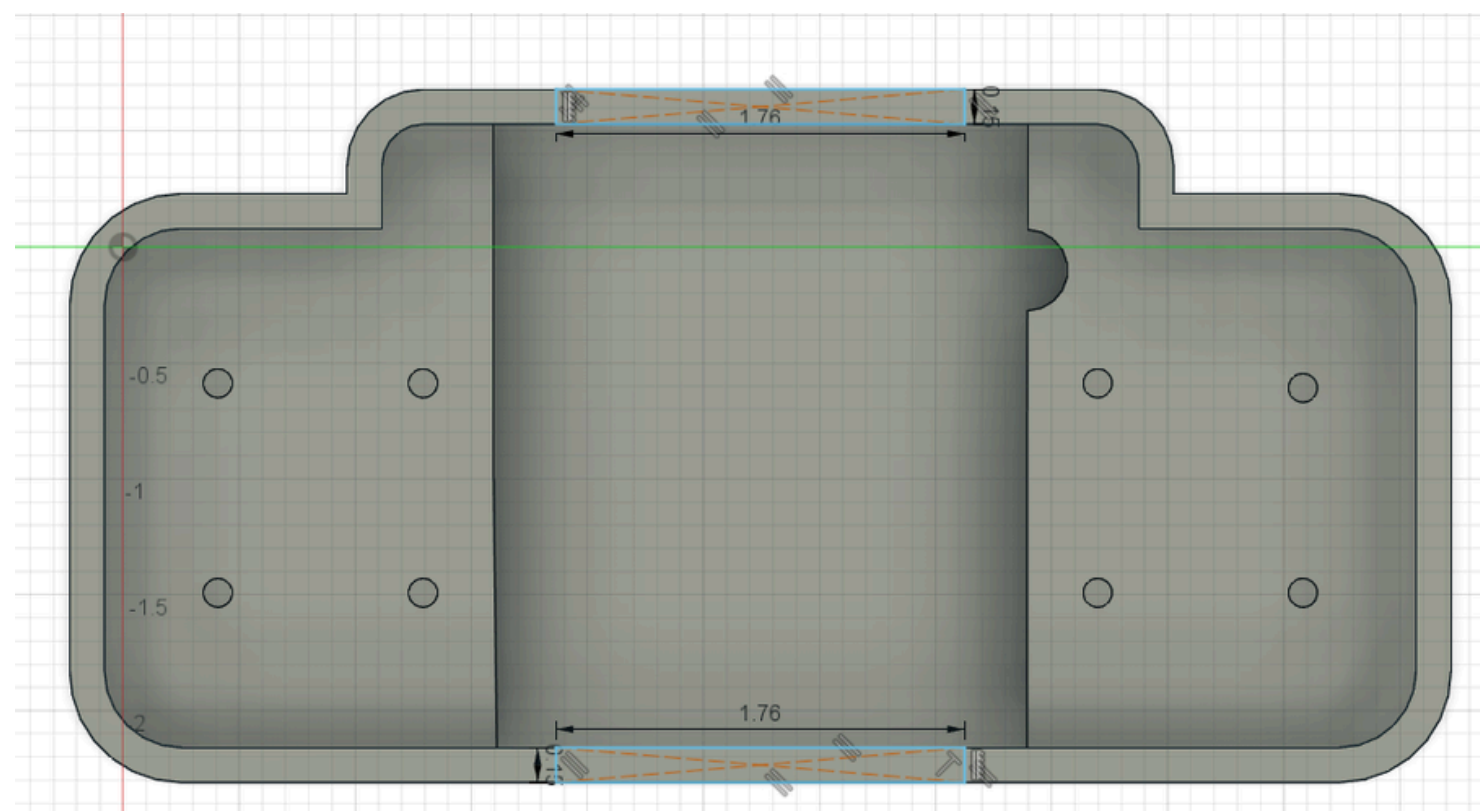
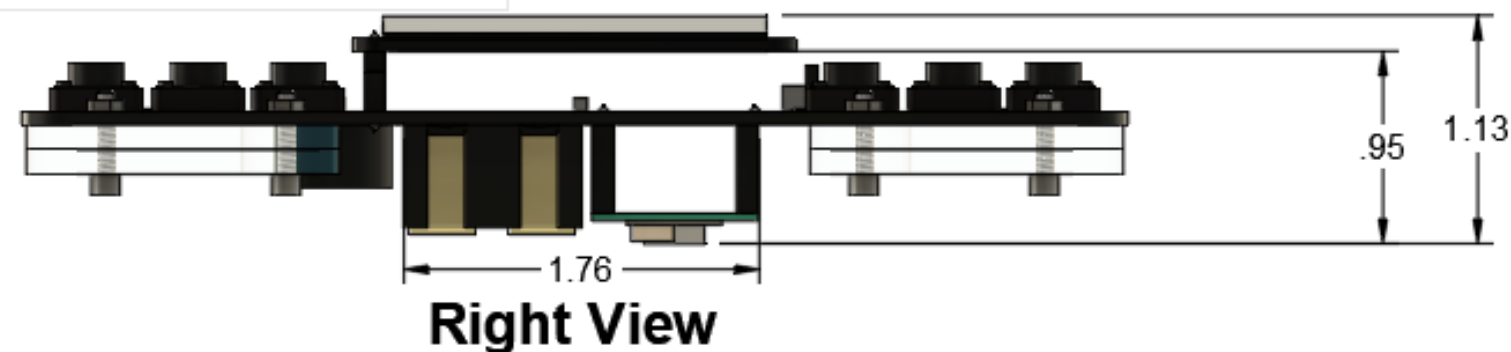


Accounting for the bolt heads is a very time-consuming task, so I offered to provide you with the layout on the left. That being said, manually finding the positions of the bolt heads would be the same as finding the position of the circular component. We would have to measure each head's position from the length and width of the sprig, use the dimension tool to reposition them, and create an indent that would allow the bolt head to slide into. After you finish your sketch, extrude each circle downward by 0.115".



# Adding Handles

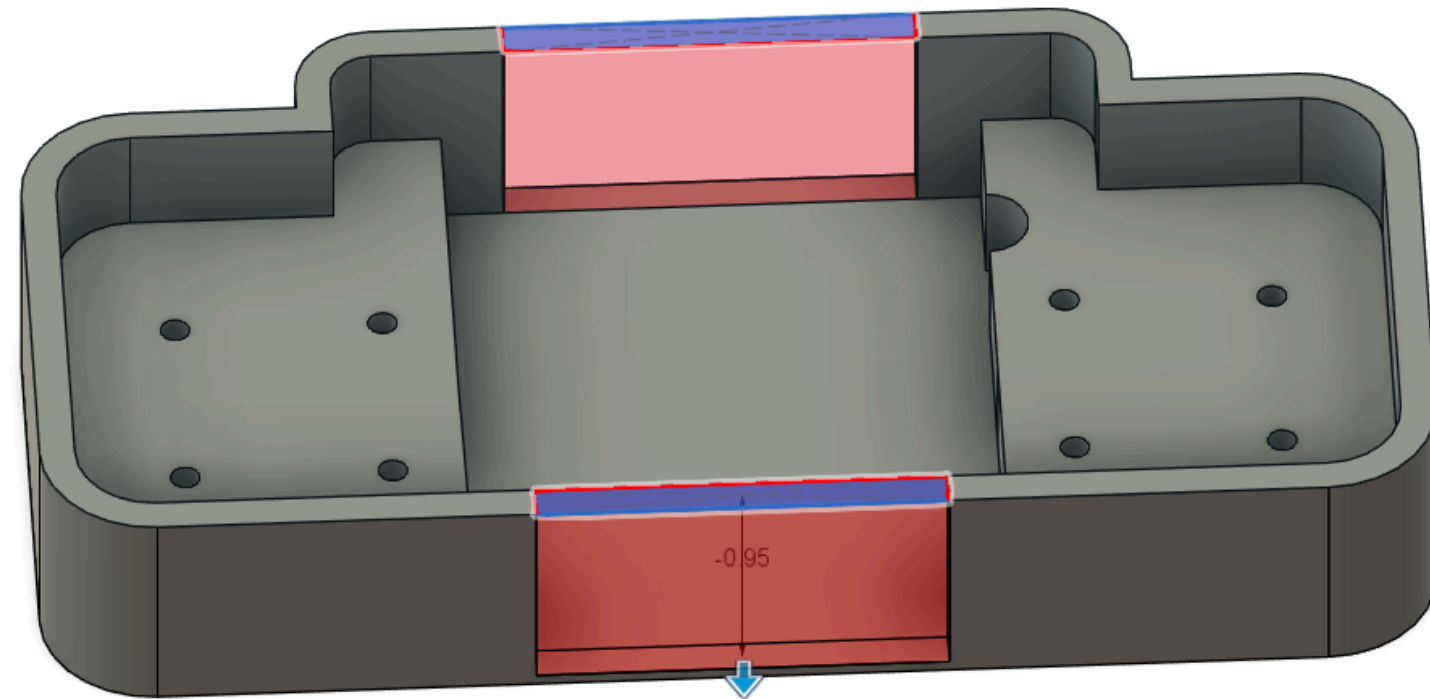
Select edge or specify point on an object



Let's finally deal with the port on the circuit board. Rather than make a small cutout only for the port, we can make a larger cutout that will also help us easily remove the sprig from the enclosure. Looking at our drawing, we see that the distance between the battery holder and circuit board is 1.76". On the top face of our enclosure, find the mid point at the top and bottom and create two 1.76" x 0.15" center rectangles. Center rectangles is under the rectangle drop down in the sketch menu. You may need to make placeholder lines to make the center rectangles.



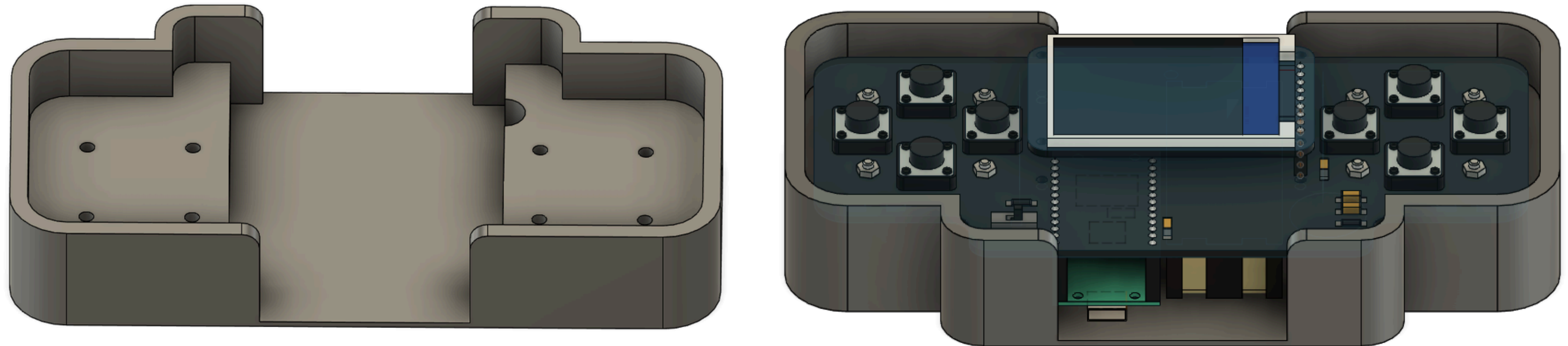
# Finishing Handles



Finally, we can extrude our sectors down by 0.95" to make the necessary cutouts for our handles. We can finish by filleting the rougher edges by 0.15" to make the design look more smooth.



# Progress Check #3



Well done, we're finally finished with the basic enclosure design. Here's a side-by-side comparison with and without the sprig.



# Mandatory Additions

This guide was meant to prepare you to take the basic enclosure taught and add your own twist to it!

This can include, but is not limited to:

- A screwable lid for the enclosure that covers up everything except the screen, buttons, and switch
- Extra designs, like flowers and textures
- An Xbox style handle
- A carving with your name on it





# Conclusion

You are now ready to design your own CAD enclosure for the Sprig board! Remember: to submit this project for tokens and rewards, you

**MUST** add a considerable addition to the enclosure to show that you took the basic design to the next level.

Have fun, and good luck!



# ENCLOSURE

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Design a custom enclosure for your project, get a 3D print of it shipped to you, and win other prizes!

